

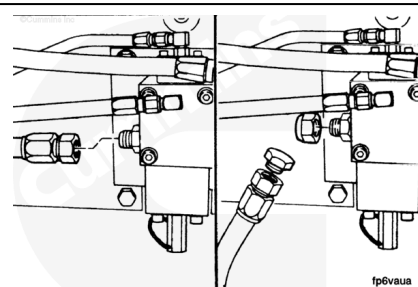
Trainings ▾

General Information

with Mechanically Actuated Injector

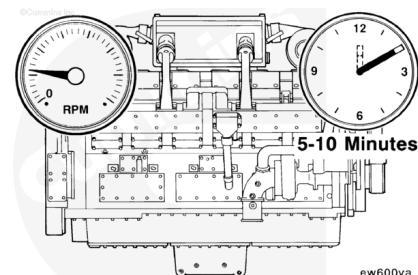
The Outer Base Circle Method overhead setting procedure can be used for fixed time (PT®) and Hydraulic Variable Timing/STC injectors, regardless of CPL or build date.

On engines with a hydro-mechanical STC valve, remove the oil supply from the oil control valve. Plug the hose, cap, and fitting. This prevents the engine from going into advance timing.



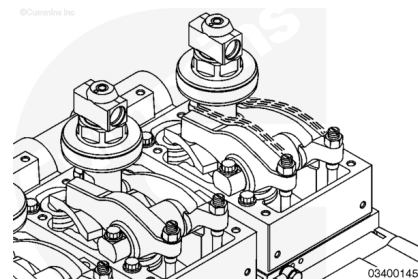
Operate the engine at high idle for 5 minutes in normal timing mode. This will allow all the lubricating oil to pump out of the injector tappets so a correct injector adjustment can be made.

Shut the engine OFF.



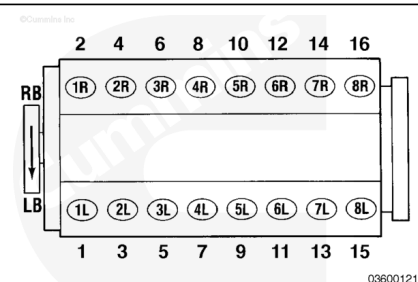
with Electronically Actuated Injector

For engines with electronically actuated injectors, periodic valve adjustment is **not** required. It is recommended that the valves be adjusted **only** when an injector is removed.



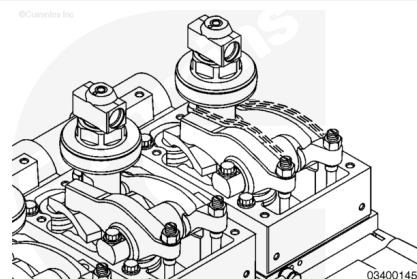
The cylinders are numbered from the front gear cover end of the engine.

To determine the right and left bank on the QSK38 and QSK50 engine, stand at the rear of the engine and face the front.

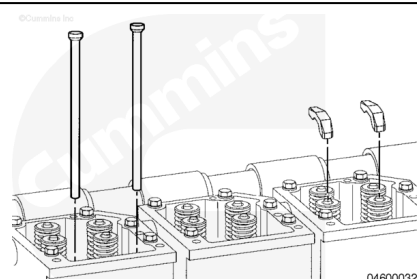


When referring to system electronics, a standard cylinder numbering system has been adopted. This is **only** used for the electronic system components on engines with electronically actuated injectors.

Each cylinder has two rocker levers. When facing the cylinder head, the rocker lever on the left is the exhaust rocker lever. The rocker lever on the right is the intake rocker lever.



If the push rods are removed for the service, make sure they are installed in the same locations.

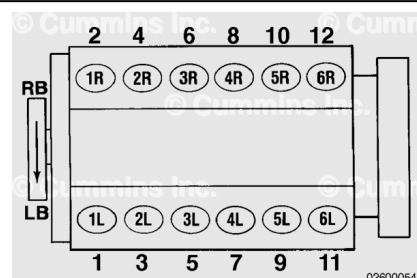


QSK38 Firing Order:

1R-6L-5R-2L-3R-4L-6R-1L-2R-5L-4R-3L or 2-11-10-3-6-7-12-1-4-9-8-5.

RB = Right Bank of cylinders.

LB = Left Bank of cylinders.



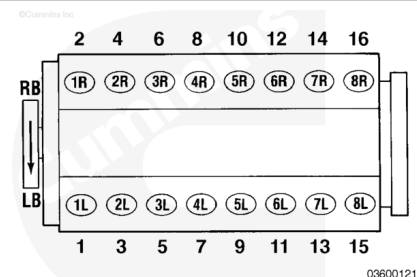
QSK50 Firing Order:

1R-1L-3R-3L-2R-2L-5R-4L-8R-8L-6R-6L-7R-7L-4R-5L or 2-1-6-5-4-3-10-7-16-15-12-11-14-13-8-9

RB = Right Bank of cylinders.

LB = Left Bank of cylinders.

The normal rotation direction for QSK38 and QSK50 is **clockwise**, when viewed from the front of the engine.



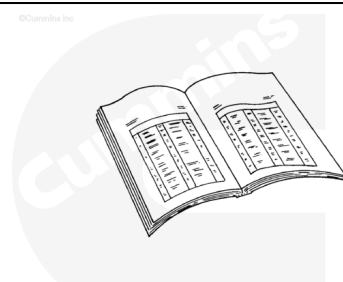
Preparatory Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

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ck800wa

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

- Disconnect the battery cables. Refer to the OEM service manual.
- Steam clean the engine to prevent dirt from entering the engine when the valve covers are removed. Refer to Procedure 000-009 in Section 7.
(/qs3/pubsys2/xml/en/procedures/99/99-000-009.html)
- Crankshaft breathers can be mounted in any cylinder location. Record the number of breathers and their correct location before removal.
- Remove the rocker lever covers and discard the gasket. Refer to Procedure 003-011
(/qs3/pubsys2/xml/en/procedures/28/28-003-011-om.html) in Section A.

with Electronically Actuated Injector

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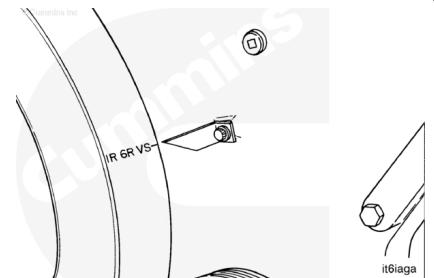
- Disconnect the battery cables. Refer to the OEM service manual.
- Steam clean the engine to prevent dirt from entering the engine when the valve covers are removed. Refer to Procedure 000-009 in Section 7. (</qs3/pubsys2/xml/en/procedures/99/99-000-009.html>)
- Crankshaft breathers can be mounted in any cylinder location. Record the number of breathers and their correct location before removal. Refer to Procedure 003-019 in Section 10. (</qs3/pubsys2/xml/en/procedures/28/28-003-019-om.html>)
- Remove the rocker lever covers and discard the gasket. Refer to Procedure 003-011 (</qs3/pubsys2/xml/en/procedures/28/28-003-011-om.html>) in Section A.

Adjust

with Mechanically Actuated Injector

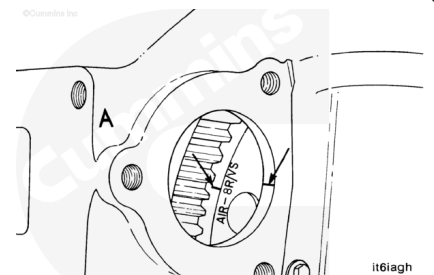
Valve and injector adjustment marks are in three locations.

Valve and injector adjustment marks are on the vibration damper. The marks **must** be aligned with the pointer.



⚠ CAUTION ⚠

When using the starter covers on the right bank, the marks on the flywheel that begin with an "A" must be used or the valves and injectors will not be adjusted correctly, causing damage to the push rods.

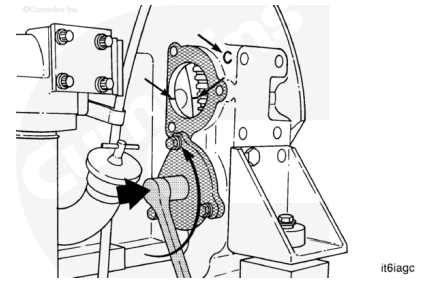


For valve and injector adjustment marks on the flywheel, with the engine barring device located on the right bank of the engine:

- The starter bore cover **must** be removed to see the marks.

⚠ CAUTION ⚠

When using the starter covers on the left bank, the marks on the flywheel that begin with a “C” must be used or the valves and injectors will not be adjusted correctly, causing damage to the push rods.



For valve and injector adjustment marks on the flywheel, with the engine barring device located on the left bank of the engine:

- The starter bore cover **must** be removed to see the marks.

This illustration also shows the engine barring device. To use the device, remove the clip and push the device shaft toward the flywheel. The barring device **must** be rotated **counterclockwise** to turn the flywheel and crankshaft in the direction of normal rotation.

The following is the firing order for all K38 engines.

- 1R-6L-5R-2L-3R-4L-6R-1L-2R-5L-4R-3L

Note : Not all K50 engines have the same firing order. Some engines manufactured after September 1986 have a revised firing order. These engines have decals on the rocker lever covers and the engine dataplate is stamped with “REVISED FIRING ORDER”.

The following is the standard firing order for all K50 engines which were manufactured before September 1986.

- 1R-1L-3R-3L-7R-7L-5R-5L-8R-8L-6R-6L-2R-2L-4R-4L

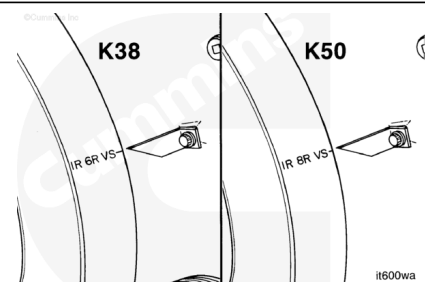
The following is the “REVISED FIRING ORDER” used on some engines built after September, 1986.

- 1R-1L-3R-3L-2R-2L-5R-4L-8R-8L-6R-6L-7R-7L-4R-5L

Two full engine rotations are required to adjust the valves and injectors on K38 and K50 engines.

The normal rotation direction for K38 and K50 engines is **clockwise**, when viewed from the front of the engine.

The valve set is represented by “VS”. Ignore the “TC” (top center) marks while setting the valves and injectors.



Determine the cylinder in position for valve set:

- The crossheads and valves are ready to be adjusted on the cylinder that has all the valves closed
- Check the two cylinders shown on the "VS" mark.

Note : The KTA50-G3, KTA50-G4, KTTA50-G2, K2000E, K1800E, and K1500E engines contain a unique camshaft that creates a noticeable difference in the height of the valve adjusting screws. When the valves are properly adjusted on these engines, the exhaust valve adjusting screw will have approximately one thread visible above the top of the lock nut. The intake valve adjusting screw will have approximately three threads visible above the top of the adjusting screw.

If the rocker lever assemblies have been removed, use this step to determine the cylinder to be set.

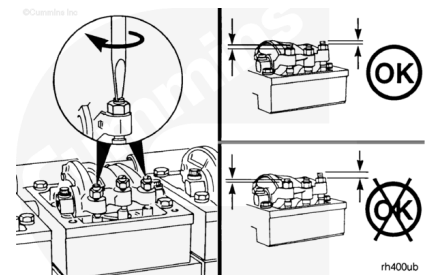
All adjusting screws **must** be loose on all cylinders, and the push rod **must** remain in alignment.

Perform this step on both cylinders to be checked.

Hold both rocker levers against the crossheads. Turn the adjusting screws until they touch the push rods. Turn the lock nuts until they touch the levers.

The cylinder with the adjusting screws that are nearly the same height (intake screw can be up to two threads above the exhaust) is ready for valve adjustment. The second cylinder that is **not** ready for adjustment will have the adjusting screw for the exhaust valve more than five threads above the intake screw.

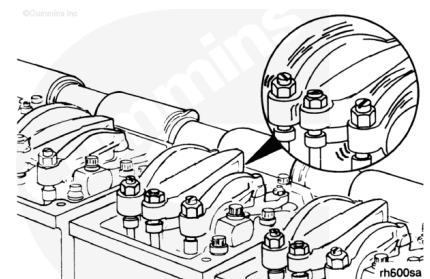
The push rods will be close to the same height above the top of the rocker lever housing on the cylinder ready for valve adjustment.



⚠ CAUTION ⚠

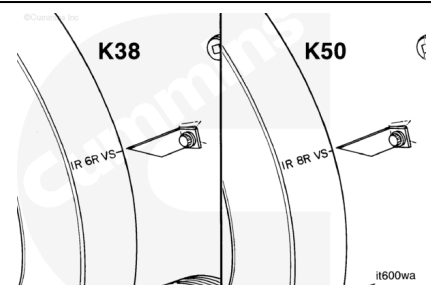
Use the correct chart for the engine being serviced or the parts can be damaged.

If the rocker levers have **not** been removed, wiggle the valve rocker levers on the two cylinders in question. The crossheads and valves on the cylinder where both levers feel loose are ready to adjust.



After identifying the cylinder with the valves ready to be adjusted, use the following charts for the sequence. The steps and specifications for adjusting the crossheads, valves, and injectors follow the charts.

Adjustment can begin on any valve set mark. In the illustration, assume the 1R-6R or 1R-8R marks are aligned and the adjusting screw height for the valves on the cylinder number 1 right-bank are closed and ready to adjust.



The following charts give the crosshead, valve, and injector adjustment sequence.

K38 (Outer Base Circle) Valve and Injector Adjustment Chart

	Valves Closed	Adjust Valves	Adjust Injectors
Valve Set Mark "VS" (L - Left-Bank) (R- Right-Bank)	On Cylinder Number (L - Left-Bank) (R- Right-Bank)	On Cylinder Number (L - Left-Bank) (R- Right-Bank)	On Cylinder Number (L - Left-Bank) (R- Right-Bank)
1R-6R VS	1R	1R	2R
6L-1L VS	6L	6L	5L
5R-2R VS	5R	5R	4R
2L-5L VS	2L	2L	3L
3R-4R VS	3R	3R	1R
4L-3L VS	4L	4L	6L
1R-6R VS	6R	6R	5R
6L-1L VS	1L	1L	2L
5R-2R VS	2R	2R	3R
2L-5L VS	5L	5L	4L
3R-4R VS	4R	4R	6R
4L-3L VS	3L	3L	1L

K50 (Outer Base Circle) Valve and Injector Adjustment Chart

	Valves Closed	Adjust Valves	Adjust Injectors
Valve Set Mark "VS" (L - Left-Bank) (R- Right-Bank)	On Cylinder Number (L - Left-Bank) (R- Right-Bank)	On Cylinder Number (L - Left-Bank) (R- Right-Bank)	On Cylinder Number (L - Left-Bank) (R- Right-Bank)
1R-8R VS	1R	1R	6R
1L-8L VS	1L	1L	6L
3R-6R VS	3R	3R	2R
3L-6L VS	3L	3L	2L

K50 (Outer Base Circle) Valve and Injector Adjustment Chart			
	Valves Closed	Adjust Valves	Adjust Injectors
2R-7R VS	7R	7R	4R
2L-7L VS	7L	7L	4L
4R-5R VS	5R	5R	1R
4L-5L VS	5L	5L	1L
1R-8R VS	8R	8R	3R
1L-8L VS	8L	8L	3L
3R-6R VS	6R	6R	7R
3L-6L VS	6L	6L	7L
2R-7R VS	2R	2R	5R
2L-7L VS	2L	2L	5L
4R-5R VS	4R	4R	8R
4L-5L VS	4L	4L	8L

Note : For K50 STC and Hydraulic Variable Timing engines, it is important to know if the engine has the standard firing order or the revised firing order. Do not use the standard firing order sequence for uprate engines manufactured after September, 1986, that have the revised firing order. All engines that have the "REVISED FIRING ORDER" have STC or Hydraulic Variable Timing injectors and are identified as revised on the engine dataplate. These engines also have decals on the rocker lever covers.

K50 (Outer Base Circle) Revised Firing Order with STC or Hydraulic Variable Timing Valve and Injector Adjustment Chart			
	Valves Closed	Adjust Valves	Adjust Injectors
Valve Set Mark "VS" (L - Left-Bank) (R- Right-Bank)	On Cylinder Number (L - Left-Bank) (R- Right-Bank)	On Cylinder Number (L - Left-Bank) (R- Right-Bank)	On Cylinder Number (L - Left-Bank) (R- Right-Bank)
1R-8R VS	1R	1R	6R
1L-8L VS	1L	1L	6L
3R-6R VS	3R	3R	7R
3L-6L VS	3L	3L	7L
2R-7R VS	2R	2R	4R
2L-7L VS	2L	2L	5L
4R-5R VS	5R	5R	1R
4L-5L VS	4L	4L	1L
1R-8R VS	8R	8R	3R
1L-8L VS	8L	8L	3L
3R-6R VS	6R	6R	2R
3L-6L VS	6L	6L	2L
2R-7R VS	7R	7R	5R

K50 (Outer Base Circle) Revised Firing Order with STC or Hydraulic Variable Timing Valve and Injector Adjustment Chart

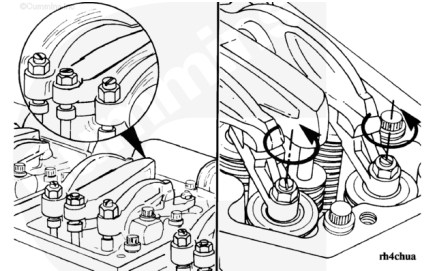
	Valves Closed	Adjust Valves	Adjust Injectors
2L-7L VS	7L	7L	4L
4R-5R VS	4R	4R	8R
4L-5L VS	5L	5L	8L

Note : Some engines have guideless crossheads and do not require adjustment.

Note : Crosshead adjustment must always be made before attempting to adjust the valves.

Adjust the crossheads on the cylinder that has both valves closed.

Loosen the crosshead adjusting screw lock nuts on the intake and exhaust valve crossheads.

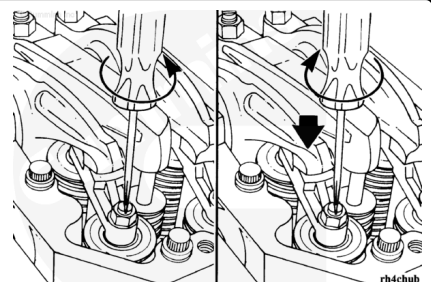


Use the following procedure to adjust both the intake and exhaust crossheads.

Turn the adjusting screw out at least one turn.

Hold the crosshead down against its guide.

Turn the adjusting screw in until it touches the top of the valve stem, but does **not** raise the crosshead.



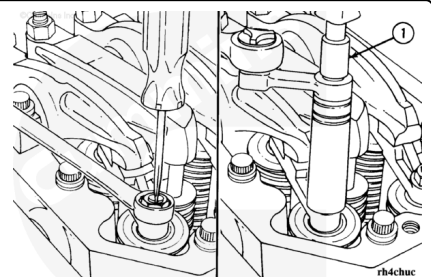
Hold the adjusting screw in this position. The adjusting screw **must not** turn when the lock nut is tightened to its torque value. Tighten the lock nut. The following torque values are given with and without torque wrench adapter (1), Part Number 3163196.

Torque Value:

With Adapter 35 n•m [26 ft-lb]

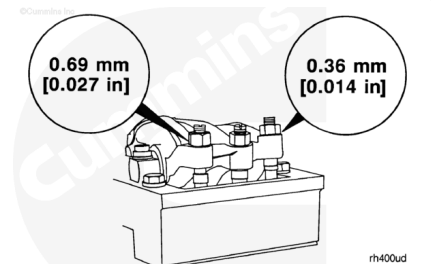
Torque Value:

Without Adapter 40 n•m [30 ft-lb]

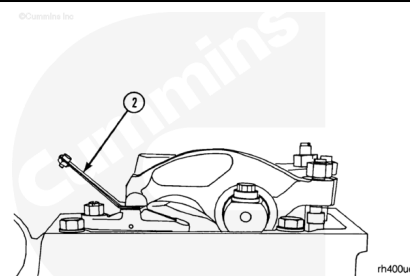


Valve Adjustment

	mm	in
Exhaust	0.69	0.027
Intake	0.36	0.014

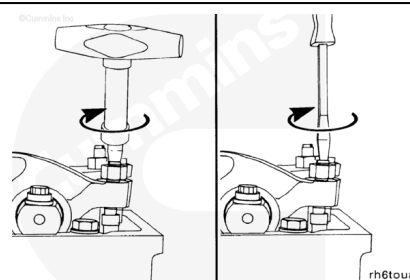


Select a feeler gauge for the correct valve lash specification. Use service tool, Part Number 3163171 (intake) or Part Number 3163172 (exhaust). Insert the gauge (2) between the rocker lever and the crosshead.



Two different methods for establishing valve lash clearance are described below. Either method can be used; however, the torque wrench method has proven to be the most consistent.

- Torque Wrench Method: Use an inch pound torque wrench, Part Number 3376592, and tighten the adjusting screw.



Torque Value: 0.68 N•m [6 in-lb]

- Feeler Gauge Method: Use a screwdriver and turn the adjusting screw **only** until the lever touches the feeler gauge.

The adjusting screw **must not** turn when the lock nut is tightened.

Tighten the lock nut to the value indicated below.

Torque Value:

With Adapter 45 N•m [33 ft-lb]

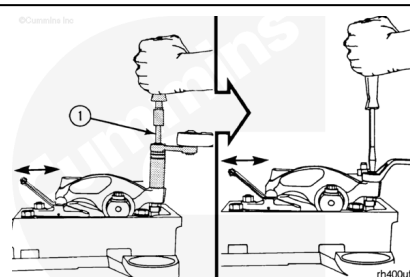
Torque Value:

Without Adapter 60 N•m [44 ft-lb]

The feeler gauge **must** slide backward and forward with **only** a small amount of drag.

Attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker. The valve lash is **not** correct when the thicker gauge will fit.

Repeat the adjustment process until the clearance is correct on both the intake and exhaust valves for the cylinder being adjusted.

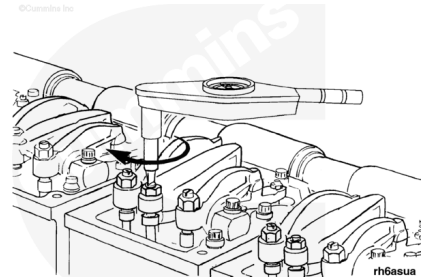


Note : The torque wrench **must** be calibrated, have a resolution of 0.28 N•m [2.5 in-lb], and have a range of 17 to 23 N•m [150 to 204 in-lb]. Do not use a clicker-type torque wrench.

Use a dial type torque wrench to tighten the injector rocker lever adjusting screw. If the screw causes chattering during setting, repair the screw and lever as required.

Hold the torque wrench in a position that allows you to look in a direct line at the dial. This is to make sure the dial will be read accurately.

Tighten the adjusting screw to make sure the parts are in alignment and to squeeze the oil out of the valve train.



Torque Value: 11 n•m [97 in-lb]

Loosen the adjusting screw at least one turn.

Tighten the adjusting screw.

Torque Value: 10 n•m [89 in-lb]

Hold the adjusting screw in this position. The adjusting screws **must not** turn when the lock nut is tightened.

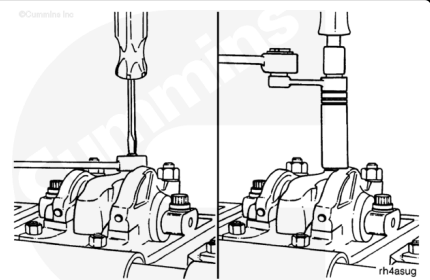
Tighten the lock nut.

Torque Value:

With Adapter 45 n•m [33 ft-lb]

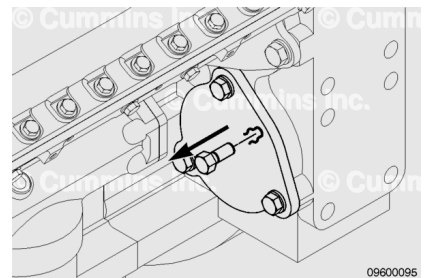
Torque Value:

Without Adapter 60 n•m [44 ft-lb]



If the barring device was used, allow the spring to push the shaft and clear the ring gear.

Install the clip.

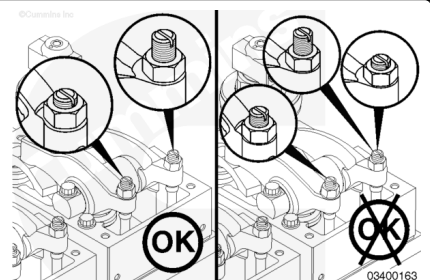


with Electronically Actuated Injector

Note : The following procedure applies to the QSK38 MCRS engine.

If the rocker lever assemblies have been removed, complete the following list on both cylinders:

- Lubricate the adjusting screw threads with clean engine oil prior to making valve and injector adjustments.
- All adjusting screws **must** be loose on all cylinders, and the pushrods **must** remain in alignment.



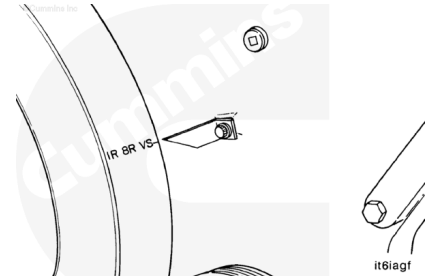
- Hold both rocker levers against the crossheads. Turn the adjusting screws until they touch the pushrods. Turn the locknuts until they touch the levers.
- The pushrods will be close to the same height above the top of the rocker lever housing on the cylinder that is ready for adjustment.
- The number of threads visible above the adjusting nut will **not** be the same. There will be more threads visible on the intake adjusting screw than on the exhaust.

The QSK38 engine has valve adjustment marks on the vibration damper and on both sides of the flywheel housing.

Valve adjustment marks **must** be aligned with the pointer. If **not**, a false adjustment can occur.

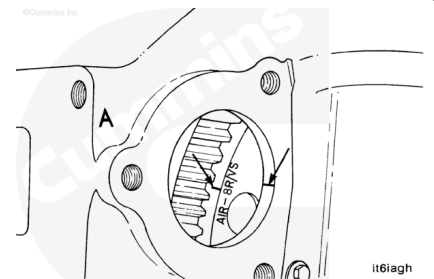
One pair of valves is adjusted at each pulley index mark, before rotating the engine to the next index mark.

Two crankshaft revolutions are required to adjust all of the valves.



⚠ CAUTION ⚠

When using the starter covers on the right bank, the marks on the flywheel that begin with an A must be used or the valves will not be adjusted correctly, causing damage to the engine.



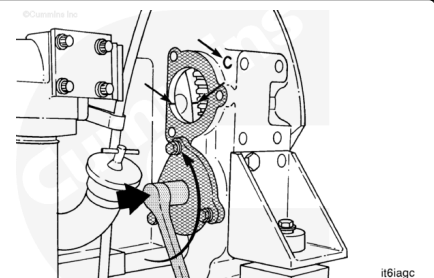
When using the valve adjustment marks on the flywheel, the upper starter bore cover **must** be removed to see the marks.

The A marks **must** be used if the marks are viewed on the right bank.

The C marks **must** be used if the marks are viewed on the left bank.

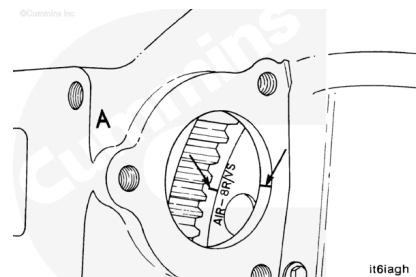
⚠ CAUTION ⚠

When using the starter covers on the left bank, the marks on the flywheel that begin with a C must be used or the valves will not be adjusted correctly, causing damage to the engine.



This illustration shows the engine barring device. To use the device, remove the clip and push the device shaft toward the flywheel. The barring device **must** be rotated **counterclockwise** to turn the flywheel and crankshaft in the direction of normal rotation.

VS represents the valve set. Ignore any TC (top center) marks while setting the valves.



Determine the cylinder in position for valve set:

The valves are ready to be adjusted on the cylinder that has all of the valves closed.

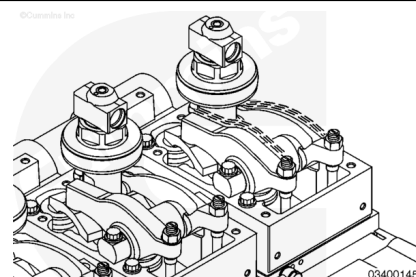
Note : When all of the valves are closed, there is some movement in the rocker levers.

Check the two cylinders shown on the VS mark.

If the rocker levers have **not** been removed, wiggle the valve rocker levers on the two cylinders in question. Set the valves on the cylinder where both levers feel loose.

Adjustment can begin on any valve set mark.

Use the following chart as a worksheet to keep track of where you are in the valve set procedure.



QSK38 Outer Base Circle Valve Set

VS Mark	Valves Closed on Cylinder Number	Adjust Valves on Cylinder Number
1R-6R VS	1R (2)	1R (2)
6L-1L VS	6L (11)	6L (11)
5R-2R VS	5R (10)	5R (10)
2L-5L VS	2L (3)	2L (3)
3R-4R VS	3R (6)	3R (6)
4L-3L VS	4L (7)	4L (7)
1R-6R VS	6R (12)	6R (12)
6L-1L VS	1L (1)	1L (1)
5R-2R VS	2R (4)	2R (4)

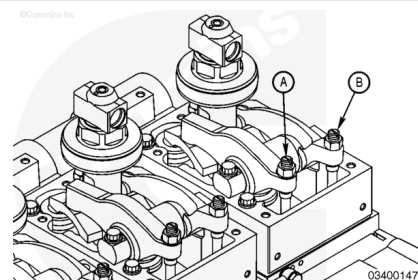
QSK38 Outer Base Circle Valve Set

VS Mark	Valves Closed on Cylinder Number	Adjust Valves on Cylinder Number
2L-5L VS	5L (9)	5L (9)
3R-4R VS	4R (8)	4R (8)
4L-3L VS	3L (5)	3L (5)

Valve Clearances

	mm		in
Exhaust Valves (A)	0.69	MAX	0.027
Intake Valves (B)	0.36	MAX	0.014

There are two different methods used to set valve lash clearance: Torque Wrench Method and Feeler Gauge Method; both are described below. Either method can be used. However, the torque wrench method has proven to be the most consistent.

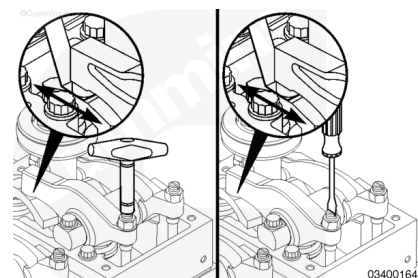


Make sure the crossheads are firmly in place on the valve stems.

Select the proper feeler gauge for the valves being set. Use service tool, Part Number 3163171 (intake) or 3163172 (exhaust).

Make sure the feeler gauge is under the center of the ball-and-socket, or the socket can rock or tip, resulting in an incorrect adjustment. Hold the swivel foot flat when checking the lash to avoid false readings.

The adjustment screws **must** turn freely, or a false reading or setting can occur.



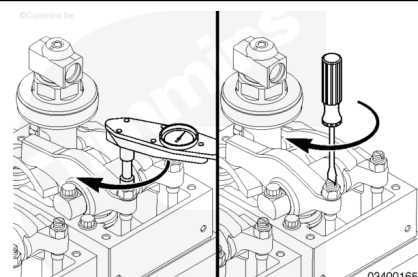
Valve Adjustment - Torque Wrench Method

Make sure the parts are aligned, and squeeze the oil out of the valve train by tightening the adjusting screw.

Loosen the adjusting screw at least one revolution.

Insert the feeler gauge between the rocker lever socket and the crosshead.

Use torque wrench, Part Number 3376592, and tighten the adjusting screw.



Torque Value: 0.68 n•m [6 in-lb]

Remove the feeler gauge.

The adjusting screw **must not** turn when the locknut is tightened. Locknut torque can be applied with or without torque wrench adapter, Part Number 3163196.

Tighten the locknut.

Torque Value:

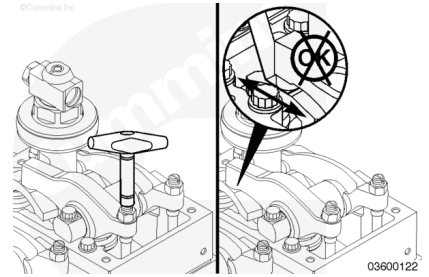
With Adapter 48 n•m [35 ft-lb]

Torque Value:

Without Adapter 60 n•m [44 ft-lb]

Attempt to insert a feeler gauge that is 0.03-mm [0.001-in] thicker. The valve lash is **not** correct if the thicker feeler gauge will fit.

Repeat the adjustment process until the proper lash is obtained.



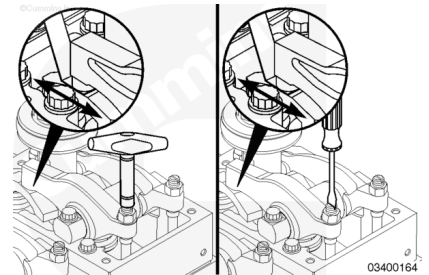
Valve Adjustment - Feeler Gauge Method

Make sure parts are aligned, and squeeze the oil out of the valve and injector train by tightening the adjusting screw.

Loosen the adjusting screw at least one revolution.

Insert the feeler gauge between the rocker lever socket and the crosshead.

Tighten the adjusting screw until the rocker lever touches the feeler gauge.



The adjusting screw **must not** turn when the locknut is tightened. Locknut torque can be applied with or without a torque wrench adapter, Part Number 3163196.

Tighten the locknut.

Torque Value:

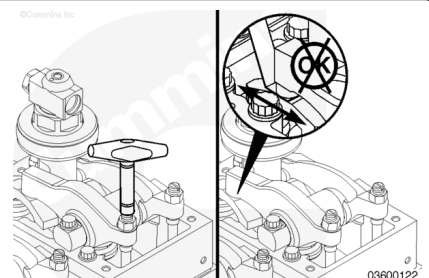
With Adapter 48 n•m [35 ft-lb]

Torque Value:

Without Adapter 60 n•m [44 ft-lb]

Attempt to insert a feeler gauge that is 0.03-mm [0.001-in] thicker. The valve lash is **not** correct if the thicker feeler gauge will fit.

Repeat the adjustment process until the proper lash is obtained.

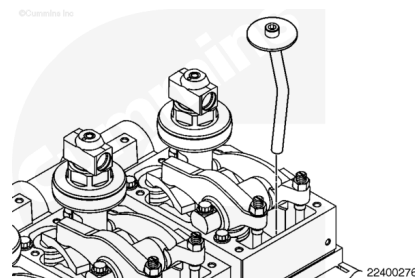


with Electronically Actuated Injector

Note : The following procedure applies to the QSK50 MCRS engine.

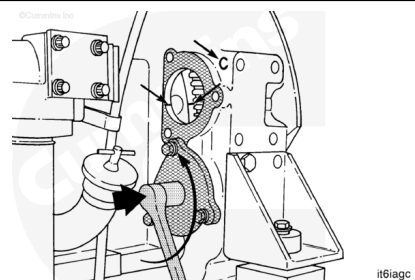
⚠ CAUTION ⚠

To reduce the possibility of engine damage, use a cylinder head protective cover to prevent tools from falling into the cam follower cavity.



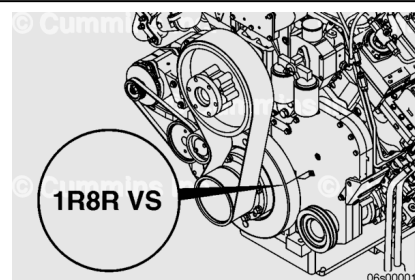
Install the cylinder head protective cover, Part Number 4918282, into the push tube hole.

This illustration shows the engine barring device. To use the device, remove the clip and push the device shaft toward the flywheel. The barring device **must** be rotated **counterclockwise** to turn the flywheel and crankshaft in the direction of normal rotation.



The 1R8R VS mark represents valve set location for cylinder 1 or 8 on the right bank. This mark will be used to set the valves on engines with electronically actuated injectors. This will allow all valves to be set in two positions.

Push the shaft in and rotate the barring device until the 1R8R VS mark on the pulley is aligned with the mark that is cast into the boss for the accessory drive seal on the front gear cover.



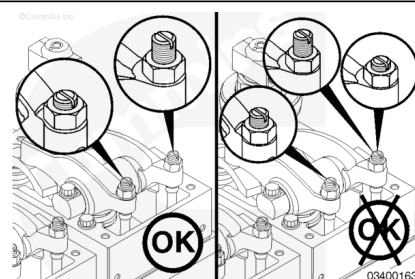
The number of threads visible above the adjusting nut will **not** be the same. There will be more threads visible on the intake adjusting screw than on the exhaust adjusting screw.

If the rocker lever assemblies have been removed, use this step to determine the cylinder to be set.

All adjusting screws **must** be loose on all cylinders, and the push rods **must** remain in alignment.

Perform this step on both cylinders to be checked.

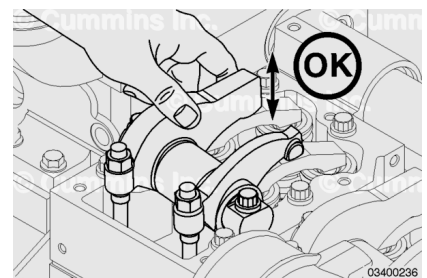
Hold both rocker levers against the crossheads. Turn the adjusting screws in until they touch the push rods. Turn the locknuts until they touch the levers.



The push rods will be the same height above the top of the rocker lever housing on the valves that should be loose.

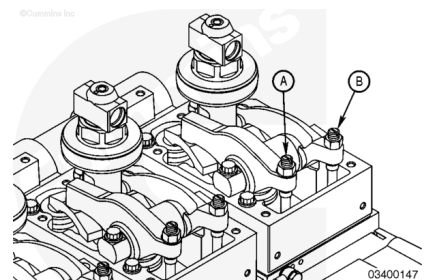
If both number 1 cylinder rocker levers on the right bank are loose, proceed to the next step. If the number 1 cylinder rocker levers on the right bank are not loose, rotate the crankshaft 360 degrees and proceed to the next step.

If the number 1 cylinder on the right bank is at 1R8R VS and both rocker levers are loose, the valve lash (overhead set) can be checked on the following rocker levers:



Intake	Exhaust
1R (2)	1R (2)
4R (8)	2R (4)
6R (12)	3R (6)
7R (14)	4R (8)
5L (9)	1L (1)
6L (11)	3L (5)
7L (13)	5L (9)
8L (15)	7L (13)

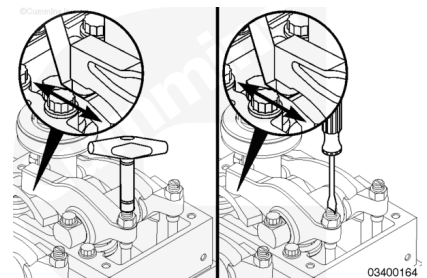
Valve Adjustment			
	mm		in
Exhaust Valves (A)	0.69	MAX	0.027
Intake Valves (B)	0.36	MAX	0.014



Use service tool, Part Number 3163171 (intake) or Part Number 3163172 (exhaust). Select a feeler gauge for the correct valve lash specification. Insert the feeler gauge between the rocker lever socket and the crosshead.

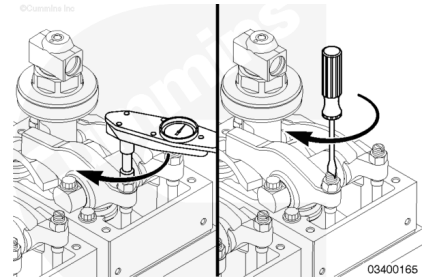
Make certain the crossheads are firmly in place on the valve stem tips.

Make certain the feeler gauge is under the center of the ball and socket or the socket can rock or tip, resulting in an incorrect adjustment. To avoid false readings, hold the swivel foot to avoid binding while checking the valve lash.



Two different methods for establishing valve lash clearance are described below:

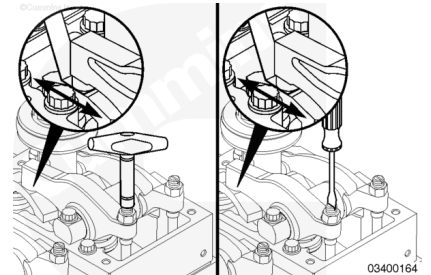
- Torque Wrench Method - use Part Number 3376592, inch-pound torque wrench to tighten the adjustment screw to 0.68 N•m [6 in-lb] torque against the feeler gauge.
- Feeler Gauge Method - use a flathead screw driver and turn the adjusting screw **only** until the lever touches the feeler gauge.



Either method can be used. The torque wrench method has proven to be the most consistent.

To set the valves using the torque wrench method, complete the following steps:

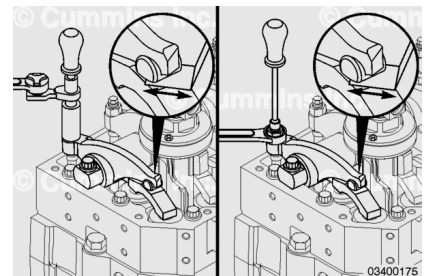
1. Be certain the parts are in alignment, Tighten the adjustment screw and squeeze the oil out of the valve train.
2. Loosen the adjustment screw at least one revolution.
3. Insert the feeler gauge between the rocker lever socket and the crosshead.
4. Use torque wrench, Part Number 3376592, and tighten the adjustment screw.
5. Remove the feeler gauge.



Torque Value: 0.68 n•m [6 in-lb]

To set the valves using the feeler gauge method, complete the following steps:

1. Make sure parts are aligned, and squeeze the oil out of the valve and injector train by tightening the adjusting screw.
2. Loosen the adjusting screw at least one revolution.
3. Insert the feeler gauge between the rocker lever socket and the crosshead.
4. Tighten the adjusting screw until the rocker lever touches the feeler gauge.



The adjustment screw **must not** turn when the locknut is tightened. Locknut torque can be applied with or without a torque wrench adapter, Part Number 3163196.

Tighten the locknut.

For the torque method (with adapter), use torque wrench adapter, Part Number ST-669.

Torque Value:

With Adapter 48 n•m [35 ft-lb]

Torque Value:

Without Adapter 60 n•m [44 ft-lb]

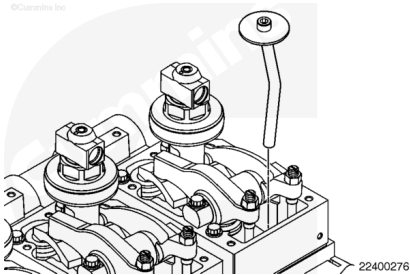
Attempt to insert a feeler gauge that is 0.03 mm [0.001 in] thicker than the valve lash specification. The lash is **not** correct when the thicker gauge will fit.

Repeat the adjustment process until the proper clearance is obtained.

Use the barring tool to rotate the crankshaft 360 degrees. Use the previous steps and specifications to set the valve lash on the following rocker levers:

Intake	Exhaust
2R (4)	5R (10)
3R (6)	6R (12)
5R (10)	7R (14)
8R (16)	8R (16)
1L (1)	2L (3)
2L (3)	4L (7)
3L (5)	6L (11)
4L (7)	8L (15)

Remove the cylinder head protective cover, Part Number 4918282, from the push tube hole.



Finishing Steps

with Mechanically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing,

remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the rocker lever covers. Refer to Procedure 003-011 in Section A. (</qs3/pubsys2/xml/en/procedures/28/28-003-011-om.html>)
- Connect the battery cables. Refer to the OEM service manual.

with Electronically Actuated Injector

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries, to reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



⚠ WARNING ⚠

Depending on the circumstance, fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.

- Install the rocker lever covers. Refer to Procedure 003-011 in Section A. (</qs3/pubsys2/xml/en/procedures/28/28-003-011-om.html>)
- Install the high pressure injector supply lines. Refer to Procedure 006-051 in Section A. (</qs3/pubsys2/xml/en/procedures/28/28-006-051.html>)
- Connect the battery cables. Refer to the OEM service manual.

